LOG-VOLUME MIDPOINT GAP AND ITS LORENTZIAN GENERALIZATION

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Abstract. This note records two statements refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Log-volume midpoint gap as a squared distance on convex bodies (Shephard, 1960) *Certificate:* Strictly Lorentzian rational cubic realized as a Steiner polynomial in $\mathbb{R}^3$; exact rational triangle-violation bound $> 7.15 \times 10^{-3}$.

Summary: Lorentzian Jensen–Bregman metric principle: $\sqrt{\mathsf{J}_G}$ a semimetric for every homogeneous Lorentzian G *Certificate:* The same cubic read as a homogeneous Lorentzian polynomial; the violation transfers verbatim.

Posed by S. Sra; counterexample found by GPT-5.6 Pro (2026-07-31); formalized by S. Sra; audited by S. Sra.

Note: I could not locate the chat but I believe GPT-5.5 Pro already broke this earlier in 2026.

For convex bodies, $K + L = \{k + \ell : k \in K,\ \ell \in L\}$ is the Minkowski sum and Vol is Lebesgue volume. Recall that d is a *semimetric* when it is symmetric, nonnegative, vanishes only on equal arguments, and satisfies the triangle inequality; the triangle inequality is the property that fails below. A homogeneous polynomial with nonnegative coefficients is *Lorentzian* in the sense of Brändén and Huh (2020) if all its second-order directional derivatives, taken at positive arguments, are quadratic forms of Lorentzian signature $(+, -, \ldots, -)$; in two variables, a form $\sum_k a_k x^k y^{d-k}$ is strictly Lorentzian exactly when the normalized coefficients $a_k/\binom{d}{k}$ are strictly log-concave, which is the criterion used below. Lorentzian polynomials are log-concave on the positive orthant, so $F = -\log G$ is convex there and every midpoint Jensen gap J_G is nonnegative. On the cone of positive definite matrices the corresponding statement is a theorem: there $\mathsf{J}_{\det}$ is the squared S-divergence, whose square root is a metric (Sra, 2016), and the same holds for a parameterized family of symmetric divergences on that cone (Sra, 2021).

Problem 1 (As posed, paraphrased: S. Sra, standalone problem posed in 2012; formulation as recorded in the audit of Appendix A)**.** The log-volume midpoint gap is the square of a distance on convex bodies: that is,

$$d_{\mathrm{Vol}}(K, L) := \left[\log \mathrm{Vol}\left(\frac{K+L}{2}\right) - \tfrac{1}{2}\log \mathrm{Vol}(K) - \tfrac{1}{2}\log \mathrm{Vol}(L)\right]^{1/2}$$

is a semimetric on convex bodies.

Problem 2 (As posed, paraphrased: S. Sra, standalone problem posed in 2019 as a Lorentzian generalization of the log-volume statement; formulation as recorded in the audit of Appendix A)**.**

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Let $\mathcal{C} \subset \mathbb{R}^m$ be an open convex cone, let $G : \mathcal{C} \to (0, \infty)$, and put $F = -\log G$ and

$$\mathsf{J}_G(x, y) := \frac{F(x) + F(y)}{2} - F\left(\frac{x+y}{2}\right).$$

If G is homogeneous and Lorentzian, then $d_G := \sqrt{\mathsf{J}_G}$ is a semimetric.

Both statements fall to a single witness: a strictly Lorentzian bivariate cubic, carried to convex bodies by Shephard's realization theorem. Throughout, $\mathsf{J}_G(x, y) = \log G(\frac{x+y}{2}) - \frac{1}{2}\log G(x) - \frac{1}{2}\log G(y)$ is the midpoint Jensen gap of $F = -\log G$, and $d_G = \sqrt{\mathsf{J}_G}$. The witness is isolated first, so that each of the two statements below is a reading of the same three numbers.

Lemma 3 (The witness)**.** *Let*

$$G(x, y) = y^3 + \tfrac{11}{5}xy^2 + \tfrac{3}{2}x^2y + \tfrac{1}{10}x^3, \qquad (x, y) \in \mathbb{R}^2_{>0}, \tag{1}$$

and take the rational points

$$p = (35, 2), \qquad q = \left(\tfrac{11}{2}, \tfrac{23}{2}\right), \qquad r = \left(\tfrac{1}{500}, 15\right). \tag{2}$$

Then G is strictly Lorentzian, and

$$d_G(p, r) - d_G(p, q) - d_G(q, r) > \frac{7156704176613847159228106087505875849130467}{10^{45}} > 0. \tag{3}$$

Proof. Writing $G = \sum_{k=0}^{3} a_k x^k y^{3-k}$ gives $(a_0, a_1, a_2, a_3) = (1, \frac{11}{5}, \frac{3}{2}, \frac{1}{10})$. By the bivariate criterion (Brändén and Huh, 2020, Example 2.3), strict Lorentzianity is strict log-concavity of $a_k/\binom{3}{k}$, and

$$\left(1, \tfrac{11}{15}, \tfrac{1}{2}, \tfrac{1}{10}\right) \text{ satisfies } \left(\tfrac{11}{15}\right)^2 = \tfrac{121}{225} > \tfrac{1}{2}, \qquad \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4} > \tfrac{11}{150}.$$

So G is strictly Lorentzian; in particular $F = -\log G$ is convex, and every Jensen gap below is nonnegative. With $R_{ab} := G((a+b)/2)^2/(G(a)G(b))$ and $\mathsf{J}_G(a, b) = \frac{1}{2}\log R_{ab}$, exact evaluation at (2) gives

$$R_{pq} = \frac{609584616081}{344696430080}, \quad R_{qr} = \frac{1254036986210090216149539001}{1235460453386242764000000000}, \quad R_{pr} = \frac{614196989908809680634186 0001}{279481339600716228000000000 0}.$$

Outward-rounded rational enclosures for the three logarithms and their square roots—no floating point—then certify (3). □

Theorem 4 ([**Disproof**]: log-volume distance on convex bodies)**.** *There are full-dimensional convex bodies $K, L \subset \mathbb{R}^3$ and Minkowski combinations A, B, C of them with*

$$d_{\mathrm{Vol}}(A, C) > d_{\mathrm{Vol}}(A, B) + d_{\mathrm{Vol}}(B, C).$$

Hence the proposed log-volume midpoint gap is not the square of a semimetric.

Proof. For convex bodies in $\mathbb{R}^3$ the relative Steiner polynomial is $\mathrm{Vol}(xK+yL) = \sum_{i=0}^{3}\binom{3}{i}W_i(K, L)x^{3-i}y^i$. Put

$$(W_0, W_1, W_2, W_3) = \left(\tfrac{1}{10}, \tfrac{1}{2}, \tfrac{11}{15}, 1\right),$$

which satisfies the full set of Shephard inequalities in dimension three:

$$W_1^2 = \tfrac{1}{4} > \tfrac{11}{150} = W_0W_2, \qquad W_1W_2 = \tfrac{11}{30} > \tfrac{1}{10} = W_0W_3, \qquad W_2^2 = \tfrac{121}{225} > \tfrac{1}{2} = W_1W_3.$$

By Shephard's realization theorem (1960, Theorem 4) these are the mixed volumes of some pair $K, L \subset \mathbb{R}^3$, whose Steiner polynomial is then exactly the cubic (1); the positive endpoint coefficients force $\mathrm{Vol}(K), \mathrm{Vol}(L) > 0$, so both bodies are full-dimensional. Setting $A = p_1K+p_2L$, $B = q_1K+q_2L$, $C = r_1K + r_2L$ for the points (2), Minkowski linearity gives $(A + B)/2 = \frac{p_1+q_1}{2}K + \frac{p_2+q_2}{2}L$ and likewise for the other pairs, so

$$d_{\mathrm{Vol}}(A, B)^2 = \mathsf{J}_G(p, q), \qquad d_{\mathrm{Vol}}(B, C)^2 = \mathsf{J}_G(q, r), \qquad d_{\mathrm{Vol}}(A, C)^2 = \mathsf{J}_G(p, r),$$

and the gap (3) of Lemma 3 transfers verbatim. □

Corollary 5 ([**Disproof**]: Lorentzian Jensen–Bregman metricity)**.** *The cubic (1) is a homogeneous strictly Lorentzian polynomial for which $d_G = \sqrt{\mathrm{J}_G}$ violates the triangle inequality on $\mathbb{R}^2_{>0}$. Hence $\sqrt{\mathrm{J}_G}$ is not a semimetric for every homogeneous Lorentzian G.*

Proof. Immediate from Lemma 3: the cubic (1) is homogeneous of degree three and strictly Lorentzian, while (3) is exactly the failure of the triangle inequality for d_G at the three points (2). □

Remark 6 (What the audit leaves standing). The midpoint integral identity and the homogeneous primitive decomposition of the earlier positive argument are correct. What fails is a claimed Gram-determinant identity for the derivative of a Hessian Rayleigh quotient, whose two sides have incompatible homogeneity in the line direction; the omitted calculation therefore conceals a false identity rather than a technical gap. Appendix A carries the full audit, including the exact enclosure procedure behind (3).

Appendix A. Two new results for hyperbolic polynomials (author: S. Sra)

These results are non-AI (by S. Sra); including here to share with interested folks.

Replacing "Lorentzian" by "hyperbolic" turns the refuted principle into a theorem. Let $p : \mathbb{R}^n \to \mathbb{R}$ be homogeneous of degree $m \geq 1$ and hyperbolic with respect to e, normalized so that $p(e) > 0$, and let Λ_{++} be the open hyperbolicity cone containing e. By Gårding (1959), Λ_{++} is an open convex cone, p is hyperbolic with respect to each of its points, and $F := p^{1/m}$ is positive, homogeneous of degree one and concave on it. The other external input is the Helton–Vinnikov theorem (Helton and Vinnikov, 2007; Lewis et al., 2005), in the normalized form: a ternary form q of degree m, hyperbolic with respect to $\mathbf{1} = (1, 1, 1)$ with $q(\mathbf{1}) > 0$, admits real symmetric A_1, A_2, A_3 with $A_1 + A_2 + A_3 = I_m$ and $q(r) = q(\mathbf{1}) \det(r_1 A_1 + r_2 A_2 + r_3 A_3)$. Three points at a time is exactly what a triangle inequality needs.

Lemma A.1 (Ternary restriction)**.** *Fix $x_1, x_2, x_3 \in \Lambda_{++}$, put $T(r) := r_1 x_1 + r_2 x_2 + r_3 x_3$ and $q := p \circ T$. Then q is homogeneous of degree m and hyperbolic with respect to $\mathbf{1}$, and each coordinate vector $\varepsilon_i \in \mathbb{R}^3$ lies in the hyperbolicity cone of q containing $\mathbf{1}$.*

Proof. Homogeneity is immediate, and $w := T(\mathbf{1}) = x_1 + x_2 + x_3 \in \Lambda_{++}$ because the cone is convex. So p is hyperbolic with respect to w, and for every $r \in \mathbb{R}^3$ the univariate $\lambda \mapsto q(r + \lambda \mathbf{1}) = p(T(r) + \lambda w)$ has only real zeros; that is hyperbolicity with respect to $\mathbf{1}$. For the second claim, the path $\gamma_i(t) = (1 - t)\varepsilon_i + t\mathbf{1}$ satisfies $T(\gamma_i(t)) = x_i + t \sum_{j \neq i} x_j \in \Lambda_{++}$, so $q(\gamma_i(t)) > 0$ on $[0, 1]$: it joins ε_i to $\mathbf{1}$ inside the positive component. □

Lemma A.2 (Positive definite representatives)**.** *In that setting the matrices A_1, A_2, A_3 supplied by Helton–Vinnikov are positive definite, and for all i, j,*

$$p(x_i) = q(\mathbf{1}) \det A_i, \qquad p\Big(\frac{x_i + x_j}{2}\Big) = q(\mathbf{1}) \det\Big(\frac{A_i + A_j}{2}\Big).$$

Proof. Along the path of Lemma A.1, $L_i(t) := \sum_k \gamma_i(t)_k A_k$ satisfies $q(\gamma_i(t)) = q(\mathbf{1}) \det L_i(t) > 0$, so $L_i(t)$ is nonsingular on $[0, 1]$; the eigenvalues of a continuous path of nonsingular real symmetric matrices cannot change sign, so the inertia is constant. At $t = 1$, $L_i(1) = A_1 + A_2 + A_3 = I_m \succ 0$, hence $A_i = L_i(0) \succ 0$. The two identities are the determinantal representation evaluated at ε_i and at $(\varepsilon_i + \varepsilon_j)/2$. □

Theorem A.3 ([**Proof**]: hyperbolic Jensen–Stein pseudometric)**.** *Let p be hyperbolic as above. Then $\sqrt{\mathrm{J}_p}$, with $\mathrm{J}_p(x, y) = \log p(\frac{x+y}{2}) - \frac{1}{2} \log p(x) - \frac{1}{2} \log p(y)$, is a pseudometric on Λ_{++}.*

Proof. Concavity of $F = p^{1/m}$ gives $F(\frac{x+y}{2}) \geq \frac{F(x)+F(y)}{2} \geq \sqrt{F(x)F(y)}$, which after raising to the mth power makes $\mathsf{J}_p \geq 0$; symmetry and vanishing on the diagonal are immediate. For the triangle inequality fix $x_1, x_2, x_3 \in \Lambda_{++}$ and take $A_1, A_2, A_3 \succ 0$ from Lemma A.2. The factor $q(\mathbf{1})$ cancels, leaving

$$\mathsf{J}_p(x_i, x_j) = \log \frac{\det((A_i + A_j)/2)}{\sqrt{\det A_i \det A_j}} = \delta_{\mathrm{S}}^2(A_i, A_j),$$

the squared Stein divergence, whose square root is a metric on the positive definite cone (Sra, 2016). Applying that triangle inequality to A_1, A_2, A_3 gives the claim, and the triple was arbitrary. □

Theorem A.4 below proves a conjecture once shared with the author (S. Sra) by Fraser (2025).

Theorem A.4 ([**Proof**])**.** *With $F = p^{1/m}$, for all a, b, c in the closed hyperbolicity cone,*

$$F(a+b)\,F(a+c) \;\geq\; F(a)\,F(a+b+c) + F(b)\,F(c).$$

Proof. Write $\mathcal{D}(M) = \det(M)^{1/m}$. We first prove the matrix form $\mathcal{D}(A+B)\mathcal{D}(A+C) \geq \mathcal{D}(A)\mathcal{D}(A+B+C) + \mathcal{D}(B)\mathcal{D}(C)$ for $A, B, C \succeq 0$. For $A \succ 0$, put $X = A^{-1/2}BA^{-1/2}$ and $Y = A^{-1/2}CA^{-1/2}$; every term carries $\mathcal{D}(A)^2$, so it suffices to show $\mathcal{D}(I+X)\mathcal{D}(I+Y) \geq \mathcal{D}(I+X+Y) + \mathcal{D}(X)\mathcal{D}(Y)$. Order the eigenvalues of X decreasingly and those of Y increasingly. Since a product does not depend on the ordering, $\mathcal{D}(I+X)\mathcal{D}(I+Y) = \left(\prod_i[(1+x_i+y_i)+x_iy_i]\right)^{1/m}$, and Minkowski's determinant inequality $\det(R+S)^{1/m} \geq \det R^{1/m} + \det S^{1/m}$ — itself Jensen's inequality for $t \mapsto \log(1+e^t)$ applied to the eigenvalues of $R^{-1/2}SR^{-1/2}$ — applied to the diagonal matrices $R = \mathrm{diag}(1+x_i+y_i)$ and $S = \mathrm{diag}(x_iy_i)$ gives

$$\mathcal{D}(I+X)\mathcal{D}(I+Y) \;\geq\; \Big(\prod_i (1+x_i+y_i)\Big)^{1/m} + \Big(\prod_i x_iy_i\Big)^{1/m}.$$

Fiedler's upper bound (Fiedler, 1971), which pairs one decreasing spectrum with the other increasing one, gives $\det(I+X+Y) \leq \prod_i(1+x_i+y_i)$, while the second term is $\mathcal{D}(X)\mathcal{D}(Y)$; a limit $\varepsilon \downarrow 0$ in $A + \varepsilon I$ covers singular A. For the cone statement, apply Lemmas A.1 and A.2 to $a, b, c \in \Lambda_{++}$ to get $A, B, C \succ 0$ with $p(ra+sb+tc) = q(\mathbf{1})\det(rA+sB+tC)$; then $F(ra+sb+tc) = q(\mathbf{1})^{1/m}\mathcal{D}(rA+sB+tC)$ for $r, s, t \geq 0$, and the displayed inequality is the matrix one times $q(\mathbf{1})^{2/m}$. Points of the closed cone are handled by $u \mapsto u + \varepsilon e$, which lies in Λ_{++}, and continuity. □

Nothing here collides with Theorem 4 and Corollary 5, and it is worth saying why not. The witness (1) is strictly Lorentzian but *not* hyperbolic: $G(x,1) = \frac{1}{10}x^3 + \frac{3}{2}x^2 + \frac{11}{5}x + 1$ has discriminant

$$18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2 = -\frac{1499}{1250} < 0 \qquad \Big(a, b, c, d = \tfrac{1}{10}, \tfrac{3}{2}, \tfrac{11}{5}, 1\Big),$$

so it has one real and two complex conjugate roots. A binary form is hyperbolic with respect to some direction only if it splits into real linear factors — each factor ℓ_i contributes the root $-\ell_i(x)/\ell_i(e)$, which is real for every real x only when ℓ_i is real up to scale — so G is hyperbolic with respect to no direction at all. In two variables hyperbolicity is therefore strictly stronger than strict Lorentzianity, and the witness lives in the gap. That gap is the content of both results together: what makes the midpoint Jensen gap a squared distance is the determinantal representation that hyperbolicity buys through Helton–Vinnikov, and not log-concavity of the coefficients.

Code and certificate

The case is the directory `counterexamples/lorentzian-jensen/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/lorentzian-jensen
python verify.py
```

- Theorem 4 (result `log-volume-distance`): exact arithmetic.
- Theorem 5 (result `lorentzian-jensen`): exact arithmetic.
- `artifacts/certificate.json`: Exact rational triangle-inequality violation bound with the three witness points.

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexamples here were found by GPT-5.6 Pro (2026-07-31). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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